AGENTX HACKATHON 2026 · TRACK PS-04

ReturnLens

Every return decision, backed by evidence.

Evidence-backed returns triage for independent online sellers.

When a customer submits an unboxing video, ReturnLens cross-references store truth, verifies deterministic code policies, and gives the solo seller a 1-click decision console.

Team Rookie (3 Members) · 24-Hour Prototype Feasibility

RETURNLENS SELLER CONSOLE — CASE #RT-8821



ORDERED: HEAD-WHT-02 (WHITE)



OBSERVED: BLACK ITEM · FRAME 00:07

P-03: COLOR CONTRADICTION DETECTED
P-04 Return Window: PASS (Day 4 of 7) | P-03 Manifest Match: CONTRADICTION FLAGGED

STATUS: ESCALATED FOR SELLER DECISION (HUMAN GATE)

Review Evidence & Authorize Draft

THE PROBLEM · SOLO SELLER BOTTLENECK

One return. Four disconnected sources.

Four systems. No traceable decision. Without a shared record, the seller connects order facts, video, and policy manually from memory.

01 BUYER MESSAGE
Subjective text & unindexed video claim in chat

02 ORDER RECORD
Shopify manifest: HEAD-WHT-02, Matte White, ₹1,499

03 COURIER STATUS
Shippo scan: delivered 4 days ago (Day 4 of 7)

04 RETURN POLICY
7-day defect window; security packaging tag rule

THE MISSING LINK

NO SHARED EVIDENCE RECORD

"What evidence supports this return decision?"

Today the seller opens 4 tabs and guesses. An erroneous refund burns inventory margin; an unfair denial destroys repeat customer trust.

The gap is not a missing return portal — it is the lack of an immutable trace linking order truth, video keyframes, and policy terms before money moves.

Fraud research: 9% strict fraud (NRF 2025) vs 15.14% fraud + abuse (Appriss/Deloitte 2024). Details in notes.

STAR DEMO · CORE BENCHMARK SCENARIO

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.



STORE CATALOG TRUTH: HEAD-WHT-02 (WHITE)



UNBOXING KEYFRAME (STAGED): MIDNIGHT BLACK

Visual Grounding Honesty: Keyframe resolution proves black casing. Exact SKU barcode is unreadable in this frame; ReturnLens factually records 'Observed Color: Black · Exact SKU: Not verified' and cites frame 00:07.

POLICY GATE EVALUATION

P-03: COLOR CONTRADICTION

STATUS: SELLER REVIEW REQUIRED

Key Findings:

- OBSERVED: Midnight Black
- UNVERIFIED: Exact SKU barcode

Seller Resolution Options:

- 1. Request unboxing box label photo
- 2. Reject return (item does not match order)
- 3. Override & approve (warehouse error)

No auto-denial · No refund without seller click

SYSTEM ARCHITECTURE · DYNAMIC AGENTIC GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Dynamic action selection and stateful resumption under deterministic human authorization.

01 MODEL ZONE

Plan & Observe

Planner validates JSON schema · Vision extracts keyframes & non-binding visual metadata.

02 CODE ZONE

Retrieve & Enforce Policy

Deterministic Python rules P-01 to P-08 · Shippo & Shopify mock APIs enforce hard rules.

03 HUMAN SELLER GATE

Authorize Consequential Action

Fail-closed authorization gate · 1-click human merchant confirmation for refund release.

HERO AGENTIC INTERACTION: DYNAMIC STATEFUL CLARIFICATION LOOP (CASE C)

Dark Frames (brightness 28) → ask_for_evidence() → WAITING_FOR_EVIDENCE → Customer Uploads → RESUME CASE → Seller Review

Why an agent? When video quality is insufficient, ReturnLens halts execution, issues a targeted re-film checklist, serializes case state into WAITING_FOR_EVIDENCE, and automatically resumes the exact case upon upload (1 retry limit prevents customer lock-out).

SECONDARY PS-04 ROUTE: Query store & courier tracking for "Where is my order?" without calling multimodal vision models.

SYSTEM OUTPUT: Immutable JSON Audit Trail recorded at every state transition

DEMO SCENARIOS · BEHAVIORAL MATRIX

Different evidence. Different action.

Predictable, controlled prototype responses across four distinct evidence conditions.

01 · CONSISTENT	**Evidence Supported · Manifest matches, packaging tag present, cosmetic defect observed**	→	**PREPARE SELLER REVIEW** Drafts mock shipping label and refund draft in pending_return_receipt. Money releases only on physical receipt.
02 · CONTRADICTION	**Ordered White, Observed Black · HEAD-WHT-02 ordered vs Frame 00:07 Black item**	→	**ESCALATE TO SELLER** Policy P-03 blocks automation. Inhibits label and refund. Seller selects: mis-shipment / box label / warehouse inspection.
03 · UNCLEAR	**Dark or Unreadable Evidence · 4s low-light video (brightness 28/255), barcode obscured**	→	**ASK AND RESUME** Calls ask_for_evidence(); transitions to WAITING_FOR_EVIDENCE; resumes context on upload (1 retry limit).
04 · SEQUENCE	**Continuity Uncertain · Staged take from sealed parcel opening to product reveal**	→	**INSPECT SEQUENCE CONTINUITY** Expected prototype behavior: inspects sequence continuity and flags cutaways for seller review.

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype buildable in one day.

00–06h · FOUNDATION

Mock APIs & Agent State

Person A (Backend)

LangGraph state machine, mock store & courier fixtures, order status route.

06–12h · EVIDENCE

Evidence Pipeline & Policy

Person B (Vision & UI)

FFmpeg keyframe indexer, vision prompt schema, rules P-01 to P-08, critic prompt.

12–18h · INTEGRATION

State Flow & Scenarios

Person C (Integration with A/B)

Clarification/resume loop, seller review console UI, wiring 4 staged demo cases.

18–24h · VALIDATION

Testing & Submission

All 3 Members

Authorization safety tests, edge cases, fallback validation, demo video recording.

MUST DELIVER (CORE PROTOTYPE)

- Deterministic policy engine (P-01 to P-08)
- Evidence-pack keyframe citations
- Fail-closed seller authorization gate
- Structured JSON audit trail log
- Stateful clarification/resume loop
- 4 reproducible demo scenarios

CUT FIRST IF BEHIND (DEFERRED EXPANSION)

- Live carrier webhooks (keep mock courier)
- Multi-item return parsing (keep 1 item/return)
- Video OCR barcode scanner (keep visual color)
- Live payment gateways (keep draft refunds)
- Merchant analytics dashboard